

21. Optical Properties of Semiconductors

21.1 Introduction

Chapter 19 made extensive use of the band theory of crystals to describe the transport properties of semiconductors. It relied upon accurate values for energy gaps, as well as upon effective masses of electrons and holes in the vicinities of band maxima and minima. However, although the electrical properties of semiconductors depend crucially upon these quantities, the dependence is indirect. The most accurate experimental determinations come from optical measurements, where photons drive electrons between various Bloch states. These measurements are not nearly so simple as they sound. Apart from the difficulties of obtaining clean surfaces and pure samples, the process of optical absorption usually demands understanding some physics beyond the simple band theory. In the case of silicon and germanium, the lowest-energy optical transitions are indirect, and they cannot proceed without aid from phonons. Even in the semiconductors with a direct optical gap, the absorption process is almost always complicated by excitons, excitations resulting from the attraction between an electron and the hole it has left behind. These observations confirm that no simple model such as the one-electron picture of energy bands in a crystal will completely capture experimental phenomena in solids and that the more precise experiments become, the more apparent will be the discrepancies.

21.2 Cyclotron Resonance

In principle, once the band structure of the semiconductor is known, so are the effective masses, because they are nothing but second derivatives of the band energy $\mathcal{E}$ with respect to the Bloch wave vector $\vec{k}$ as in Eq. (16.28). This definition provides a way to find effective masses from band structure calculations, but is almost useless if they are to be determined experimentally. Experimental determination of effective masses requires a physical probe that tests dynamics directly, a service provided in semiconductors by the technique of *cyclotron resonance*.

Effective masses enter into expressions for conductivity such as $\sigma = ne^2\tau/m^*$, but always in conjunction with the relaxation time, making it difficult to separate the two. Cyclotron resonance solves this problem by introducing a magnetic field, as shown in Figure 21.1. Classically, an electron moving at velocity v in a magnetic induction of strength B experiences a force perpendicular to the direction of motion

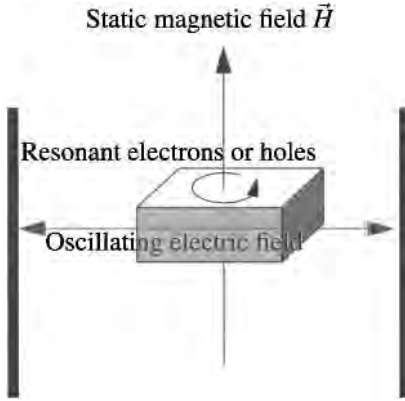


Figure 21.1. Cyclotron resonance is produced in a semiconductor by applying a static magnetic field and then applying microwave radiation at the orbital frequency of the electrons or holes.

of magnitude evB/c . Assuming it travels in a circular orbit of radius R and setting the centripetal force equal to the magnetic force

$$\frac{m^* v^2}{R} = \frac{evB}{c} \quad (21.1)$$

predicts an orbital frequency

$$\omega_c = \frac{v}{R} = \frac{eB}{m^* c} = 17.6 \frac{m}{m^*} \left[\frac{B}{\text{kG}} \right] \text{ GHz}. \quad (21.2)$$

Because this oscillation frequency is independent of the radius of the orbit, by applying a perpendicular electric field with frequency ω_c , electrons can be excited to large orbits and resonantly absorb incoming radiation. The relaxation time enters nowhere in the resonance condition, although it plays a crucial role in determining when the resonance is practically observable.

Calculation of Absorption. It is not too cumbersome to work out a detailed expression for the absorption of microwave radiation so long as the effective mass m^* of an electron (or hole) can be taken to be a scalar. In the semiclassical approximation the dynamics of electrons are given by

$$\dot{\vec{v}} + \frac{\vec{v}}{\tau} = -\frac{e\vec{E}}{m^*} - \frac{e}{m^* c} \vec{v} \times \vec{B}. \quad \tau \text{ is the relaxation time.} \quad (21.3)$$

Taking the static $\vec{B}$ field to point along the $\hat{z}$ direction, and allowing both $\vec{E}$ and $\vec{v}$ vary in time as $e^{-i\omega t}$, one has that

$$\left(-i\omega + \frac{1}{\tau} \right) \vec{v} = -\frac{e\vec{E}}{m^*} - \omega_c (\hat{x}v_y - \hat{y}v_x) \quad (21.4)$$

$$\Rightarrow \left(-i\omega + \frac{1}{\tau} \right) \vec{v} = -\frac{e\vec{E}}{m^*} - \omega_c \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \vec{v}. \quad (21.5)$$

Therefore

$$\vec{j} = -ne\vec{v} \equiv \sigma \vec{E}, \quad \vec{j} \text{ is the current density, and } n \text{ is the conduction electron density.} \quad (21.6)$$

where

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ -\sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix} \quad (21.7)$$

$$\sigma_{xx} = \frac{\sigma_0(1 - i\omega\tau)}{(1 - i\omega\tau)^2 + \omega_c^2\tau^2} \quad (21.8a)$$

$$\sigma_{xy} = -\frac{\sigma_0\tau\omega_c}{(1 - i\omega\tau)^2 + \omega_c^2\tau^2} \quad (21.8b)$$

and

$$\sigma_{zz} = \frac{\sigma_0}{1 - i\omega\tau}, \quad (21.8c)$$

with

$$\sigma_0 = \frac{ne^2\tau}{m^*}. \quad (21.8d)$$

If the applied electric field is along $\hat{x}$, then the time-averaged power absorbed by the sample, $\langle \vec{E} \cdot \vec{j} \rangle / 2$, is proportional to the real part of σ_{xx} , which is

$$\text{Re}[\sigma_{xx}] = \sigma_0 \frac{\omega_c^2\tau^2 + \omega^2\tau^2 + 1}{(\omega_c^2\tau^2 - \omega^2\tau^2 + 1)^2 + 4\omega^2\tau^2}. \quad (21.9)$$

A plot of Eq. (21.9) appears in Figure 21.2 for several values of $\omega\tau$.

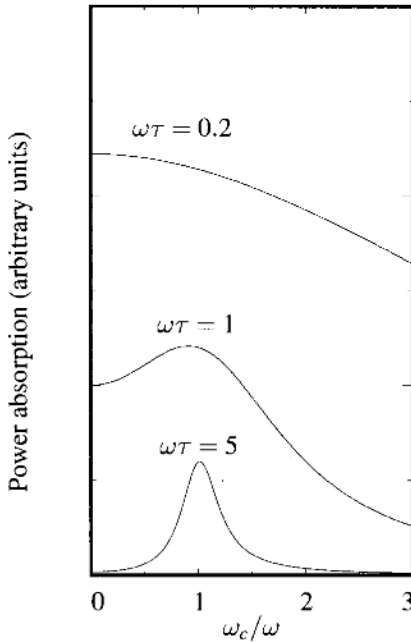


Figure 21.2. Plot of Eq. (21.9) for three values of $\omega\tau$. The frequency ω is the frequency of the oscillating electric field $\vec{E}$ —in contrast with ω_c , which is the resonant frequency of electrons governed by $\vec{B}$. When $\omega\tau \ll 1$, collisions damp the motion of the electron before it has time to complete one cycle and absorb much energy. No resonance is visible. By the time $\omega\tau = 1$ the relaxation time has become large enough to permit observation of a resonance. Varying ω_c is carried out by varying the strength of the static magnetic field, while the oscillating electric field frequency ω remains fixed.

Practical Considerations. It was a substantial achievement by Dresselhaus et al. (1953) and Lax et al. (1954) to turn cyclotron resonance into a practical tool, because for the experiment to work four competing requirements must be satisfied simultaneously.

1. One must guarantee that the semiclassical approximation be valid for the dynamics of the electrons or holes. This condition rules out optical frequencies where $\hbar\omega \gtrsim 1$ eV and interband transitions become possible.
2. The oscillating electric field must be able to penetrate the sample. This requirement is equivalent to demanding that $\omega > \omega_p$, the plasma frequency of Eq. (20.32). For undoped silicon and germanium at a temperature of 4 K, with free carrier densities on the order of 10^{13} cm^{-3} , this condition can be satisfied, but already by a temperature of 70 K the carrier densities have risen high enough that plasma oscillations mask the cyclotron resonance. This condition rules out use of this technique in metals.
3. The relaxation time must be long enough that $\omega\tau \gtrsim 1$, according to Figure 21.2, requiring (a) very pure samples to reduce impurity scattering and (b) low temperatures to avoid phonon scattering.
4. The free carrier density must be high enough that electrons and holes are available to absorb radiation. Although the density of free carriers is exponentially small at low temperatures, this problem can be overcome by irradiating samples with photons at the energy of the band gap.

21.2.1 Electron Energy Surfaces

Data from a cyclotron resonance experiment in germanium appear in Figure 21.3. There is a large number of peaks present. Because germanium is an indirect gap semiconductor, its conduction band minimum does not lie atop $\vec{k} = 0$, but instead consists of four pockets of electrons straddling the zone boundaries in the (111) direction: these electron pockets were shown in Figure 19.9(A). The constant energy surfaces in the neighborhood of the conduction band minima are not at all spherical, but instead have the form

$$\mathcal{E} = \frac{\hbar^2}{2} \left[\frac{k_1^2}{m_1^*} + \frac{k_2^2}{m_2^*} + \frac{k_3^2}{m_3^*} \right] \quad (21.10)$$

Every symmetric quadratic form can be diagonalized; the energy is being expressed here in the basis where the effective mass tensor is diagonal. In the case of germanium, k_1 is along (111), and for silicon it is along (100).

where $m_1^* = 1.64 m$ and $m_2^* = m_3^* = 0.082 m$. For arbitrary directions of the magnetic field, there is a separate peak from each pocket of electrons, although for certain symmetry directions the peaks coalesce. For silicon, the situation is similar [Figure 19.9(B)], except that the conduction band minimum lies along the line between 0 and (100), providing three rather than four distinct pockets of electrons. The effective masses for silicon are $m_1^* = 0.9 m$ and $m_2^* = m_3^* = 0.19 m$.

When the effective mass tensor is not diagonal, one must return to Eq. (21.4) to see how the resonance condition changes. Rather than finding again the shape of the absorption line, it is enough to ask for the frequencies at which resonances occur. Resonances are located by searching for solutions of Eq. (21.4) which, in

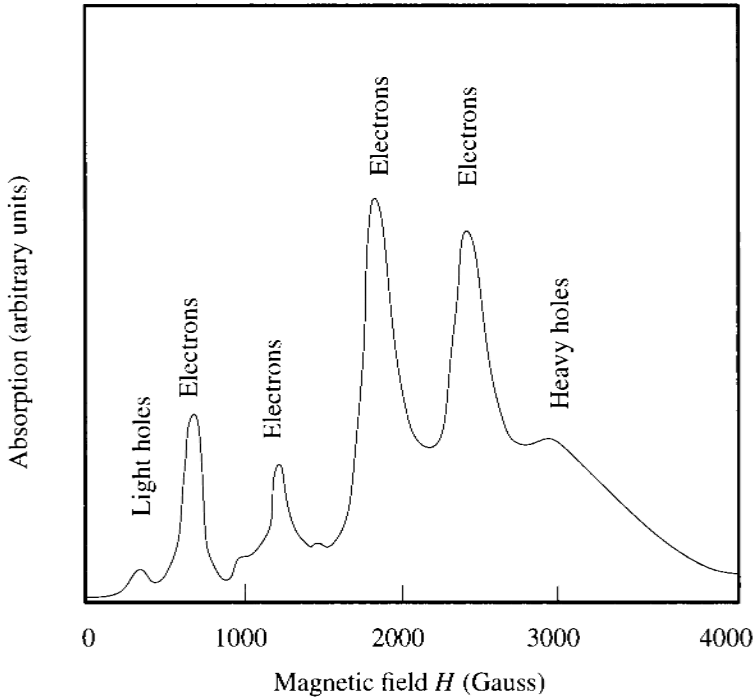


Figure 21.3. Data from cyclotron resonance in germanium. The magnetic field is oriented at 10° from the (110) plane and 30° from the [100] direction. Each of the four pockets of electrons in Figure 19.9 contributes a peak, as well as the light and the heavy holes. Because the applied electric field is not perfectly sinusoidal, but has a small frequency components of 2ω , 3ω . . . , higher harmonics of electron resonances are also visible. [Source: Dexter et al. (1956), p. 642.]

the absence of damping ($\tau = 0$), can exist when $\vec{E} = 0$. If one adopts coordinates where $\mathbf{M}$ is diagonal, then

$$0 = i\omega\vec{v} - \frac{e}{c} \begin{pmatrix} \frac{1}{m_1^*} & 0 & 0 \\ 0 & \frac{1}{m_2^*} & 0 \\ 0 & 0 & \frac{1}{m_3^*} \end{pmatrix} \begin{pmatrix} 0 & B_3 & -B_2 \\ -B_3 & 0 & B_1 \\ B_2 & -B_1 & 0 \end{pmatrix} \vec{v} \quad (21.11)$$

$$\Rightarrow \omega = \frac{e}{c} \sqrt{\sum_{\alpha=1}^3 \frac{B_\alpha^2 m_\alpha^*}{m_1^* m_2^* m_3^*}} \quad \text{Find when the determinant of the right-hand side of Eq. (21.11) vanishes.} \quad (21.12)$$

Effective masses are determined by changing the direction of the magnetic field and by observing changes in locations of the resonance peaks according to Eq. (21.12).

Hole Energy Surfaces. The valence band maxima of silicon and germanium are found at $\vec{k} = 0$. However, matters are still somewhat complicated, because these bands are twofold degenerate right at $\vec{k} = 0$, allowing two separate effective masses

for holes. The heavy hole in germanium has $m_H^* \approx 0.28 m$ and the light hole has $m_L^* \approx 0.044 m$, while the corresponding numbers for silicon are $m_H^* \approx 0.49 m$ and $m_L^* \approx 0.16 m$.

21.3 Semiconductor Band Gaps

21.3.1 Direct Transitions

Direct gap semiconductors, as defined in Section 19.3.1, have a maximum of the valence band sitting directly below the minimum energy state of the conduction band in k space.

The theory for optical transitions in this case starts with Eq. (20.70), which shows a contribution to absorption whenever the energy $\hbar\omega$ of an incoming photon equals the difference $\mathcal{E}_{n'} - \mathcal{E}_n$ between two energy states. In order to place the theoretical result in the most convenient form, it is helpful to recall Eq. (20.14) and rewrite (20.70) as

$$\text{Im}[\epsilon_{\alpha\beta}] = \frac{4e^2\pi^2}{m^2\omega^2\mathcal{V}} \sum_{ll'} (f_l - f_{l'}) \langle l | \hat{P}_\alpha | l' \rangle \langle l' | \hat{P}_\beta | l \rangle \delta(\mathcal{E}_{l'} - \mathcal{E}_l - \hbar\omega) \quad (21.13)$$

$$= \left(\frac{2\pi e}{m\omega} \right)^2 \frac{1}{\mathcal{V}} \sum_{\vec{k}n_1n_2} \langle \vec{k}n_1 | \hat{P}_\alpha | \vec{k}n_2 \rangle \langle \vec{k}n_2 | \hat{P}_\beta | \vec{k}n_1 \rangle \delta(\mathcal{E}_{n_2\vec{k}} - \mathcal{E}_{n_1\vec{k}} - \hbar\omega) \quad (21.14)$$

Labeling the states now by Bloch index $\vec{k}$ and band indices n_1 and n_2 . The Bloch indices $\vec{k}$ of the initial and final states must be the same or else the matrix element vanishes. The sum is understood to include only cases where $n_1\vec{k}$ is occupied, and $n_2\vec{k}$ is unoccupied.

$$= \left(\frac{2\pi e}{m\omega} \right)^2 |P_{\alpha\beta}(\omega)|^2 D_j(\hbar\omega), \quad (21.15)$$

where

$$|P_{\alpha\beta}(\omega)|^2 \equiv \frac{\sum_{n_1n_2\vec{k}} \langle \vec{k}n_1 | \hat{P}_\alpha | \vec{k}n_2 \rangle \langle \vec{k}n_2 | \hat{P}_\beta | \vec{k}n_1 \rangle \delta(\mathcal{E}_{n_2\vec{k}} - \mathcal{E}_{n_1\vec{k}} - \hbar\omega)}{\sum_{n_1n_2\vec{k}} \delta(\mathcal{E}_{n_2\vec{k}} - \mathcal{E}_{n_1\vec{k}} - \hbar\omega)} \quad (21.16)$$

and

$$D_j(\hbar\omega) \equiv \frac{1}{\mathcal{V}} \sum_{n_1n_2\vec{k}} \delta(\mathcal{E}_{n_2\vec{k}} - \mathcal{E}_{n_1\vec{k}} - \hbar\omega). \quad (21.17)$$

The absorption has been expressed as a product of an average matrix element $|P|$ and the *joint density of states* D_j . The reason to place the expression in this form is that the averaged matrix element has no particular reason to exhibit sharp changes as a function of frequency and can often be approximated as a constant, but the joint density of states D_j is extremely sensitive to details of the band structure and depends strongly upon frequency. The joint density of states carries all the information about whether there exist pairs of occupied and unoccupied states with the right energy separation so that the photon can excite an electron between them. It vanishes when this condition is not met. In addition, whenever the band energies $\mathcal{E}_{n_2\vec{k}}$ and $\mathcal{E}_{n_1\vec{k}}$ run parallel to one another as a function of $\vec{k}$, D_j exhibits van Hove singularities, as defined in Section 7.2.5.

For direct band gap semiconductors to which Eq. (21.14) applies, one must turn to alloys, such as GaAs, or InSb. It is natural to expect that right at the threshold of absorption, where $\hbar\omega \geq \mathcal{E}_g$, the matrix element $|P_{\alpha\beta}|$ has no reason to vary rapidly, so optical absorption should be proportional to the joint density of states $D_j(\hbar\omega)$. Because both valence and conduction bands have parabolic extrema around $\vec{k} = 0$, $\mathcal{E}_c(\vec{k}) - \mathcal{E}_v(\vec{k}) - \mathcal{E}_g$ is a parabolic function vanishing at $\vec{k} = 0$, and D_j must therefore exhibit the van Hove singularity discussed in Section 7.2.5; that is, D_j should vanish for $\hbar\omega < \mathcal{E}_g$, and then should rise as $\sqrt{\hbar\omega - \mathcal{E}_g}$.

There are few cases where such behavior is actually observed. One of them appears in Figure 21.4. More frequently, the onset of optical absorption displays the behavior shown in Figure 21.5. The expected square root shape has been overridden by an additional process, the formation of excitons, to be described in Section 21.4.

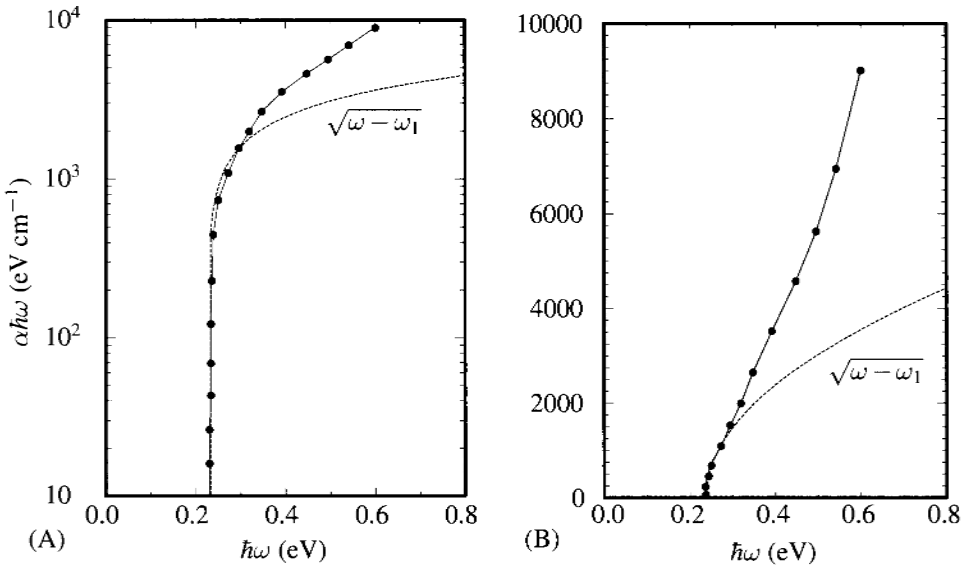


Figure 21.4. Measurement of absorption coefficient α times $\hbar\omega$, showing a van Hove singularity at onset of optical absorption in the direct gap semiconductor InSb. Data taken at a temperature of 5 K by Goebli and Fan and reported by Johnson (1967), p. 171, are compared with the expected square root form. The y axis is multiplied by $\hbar\omega$ because of the ω dependence predicted by Eq. (20.21) in combination with Eq. (21.16). (A) shows the data on a linear-log scale, while (B) shows the same data replotted on a linear scale, making the agreement look less impressive. Deviations from square root form due to variation of the matrix element $|P_{\alpha\beta}|$ produce better agreement with the experiment and were calculated by Kane (1957).

21.3.2 Indirect Transitions

As mentioned in Section 19.3.1, the elemental semiconductors germanium and silicon have indirect gaps, meaning that the conduction band minima are at different

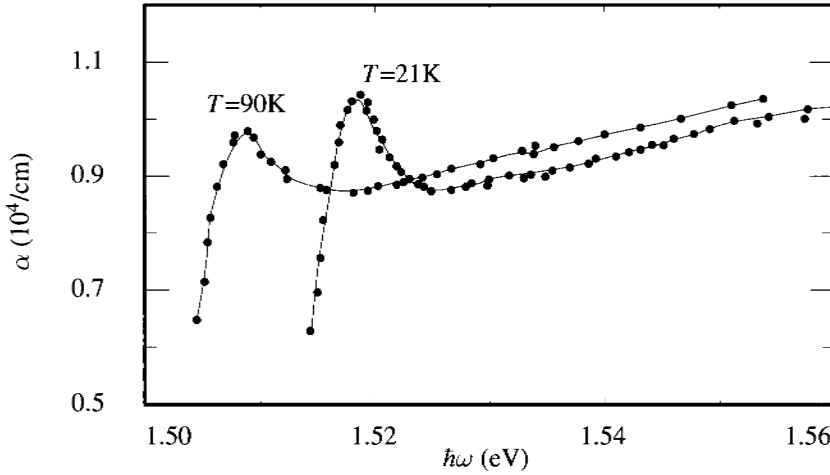


Figure 21.5. Measurement of absorption coefficient α in gallium arsenide, showing modification of absorption due to excitons. [Source: Sturge (1962), p. 771.]

locations in k space from the valence band maxima. It is possible for light to excite electrons between the valence band maximum and conduction band minimum, but only with the assistance of phonons so as to permit conservation of momentum. The microscopic theory of such transitions is therefore rather involved. The initial and final states of the system differ not only because an electron has moved to a new state, but also because the number of phonons has changed. The interaction Hamiltonian between photons and solid does not include a term for phonons, so any matrix element with the interaction Hamiltonian in the middle and with such initial and final states on either side must vanish. These transitions show up only at second order in perturbation theory. Fortunately, it is possible to guess the form of the final result without going through detailed calculations, using only conservation laws. In order for the Bloch index of an electron to change by $\vec{\delta k}$, either a phonon of wave vector $-\vec{\delta k}$ and energy $\hbar\omega_{\text{ph}}(\vec{\delta k})$ must be created, or a phonon of wave vector $\vec{\delta k}$ and energy $\hbar\omega_{\text{ph}}(\vec{\delta k})$ must be destroyed, in order to conserve crystal momentum. In order for energy to be conserved, one must have

$$\hbar\omega = \mathcal{E}_c - \mathcal{E}_v \pm \hbar\omega_{\text{ph}}(\vec{\delta k}). \quad (21.18)$$

$\mathcal{E}_c$ is the energy of an electron in the conduction band, and $\mathcal{E}_v$ is the energy of an electron in the valence band.

At temperatures much lower than the Debye temperature, there will be very few phonons residing naturally in the sample, and the negative term in Eq. (21.18) where a phonon is destroyed should cease to provide transitions. So according to Eq. (21.18), optical absorption in silicon and germanium should appear in two stages, at a first and a second frequency, with the strength of the absorption at the first frequency diminishing steadily with temperature.

The shape of the absorption curve can similarly be estimated. A complete formula will have to involve various matrix elements, sums over intermediate states,

and so on. Assuming that the energy dependence of all these can be neglected, the extinction coefficient κ should be proportional to

$$\kappa \propto \sum_{\vec{k}_c \vec{k}_v} \delta \left(\mathcal{E}_c(\vec{k}_c) - \mathcal{E}_v(\vec{k}_v) - \hbar\omega \pm \hbar\omega_{\text{ph}}(\vec{\delta k}) \right) \quad (21.19)$$

$$= \int d\mathcal{E}_c \int d\mathcal{E}_v D_c(\mathcal{E}_c) D_v(\mathcal{E}_v) \delta \left(\mathcal{E}_c - \mathcal{E}_v - \hbar\omega \pm \hbar\omega_{\text{ph}} \right) \quad (21.20)$$

Neglect the $\vec{\delta k}$ dependence of ω_{ph} .

$$\propto \int_{\mathcal{E}_g} d\mathcal{E}_c \int^0 d\mathcal{E}_v \sqrt{\mathcal{E}_c - \mathcal{E}_g} \sqrt{-\mathcal{E}_v} \delta \left(\mathcal{E}_c - \mathcal{E}_v - \hbar\omega \pm \hbar\omega_{\text{ph}} \right) \quad (21.21)$$

Letting $\mathcal{E}_g$ be the band gap, setting the valence band maximum to zero, and using the generic three-dimensional forms of the density of states near a band maximum or minimum.

$$= \int_{\mathcal{E}_g}^{\hbar\omega \mp \hbar\omega_{\text{ph}}} d\mathcal{E}_c \sqrt{\mathcal{E}_c - \mathcal{E}_g} \sqrt{\hbar\omega - \mathcal{E}_c \mp \hbar\omega_{\text{ph}}} \quad (21.22)$$

$$= (\hbar\omega \mp \hbar\omega_{\text{ph}} - \mathcal{E}_g)^2 \int_0^1 dy \sqrt{y} \sqrt{1-y}. \quad (21.23)$$

Letting $y = (\mathcal{E}_c - \mathcal{E}_g) / (\hbar\omega \mp \hbar\omega_{\text{ph}} - \mathcal{E}_g)$. From Eq. (21.22), the result vanishes, however, unless $\hbar\omega > \mathcal{E}_g \mp \hbar\omega_{\text{ph}}$. The value of the integral is not important, just the fact that it is constant.

This prediction is only in moderate accord with experiment, as shown in Figure 21.6. It is possible to pick out the two frequencies $\mathcal{E}_g \mp \hbar\omega_{\text{ph}}$. However, at low energies the curve does not have the expected form, nor does the onset of absorption at $\mathcal{E}_g + \hbar\omega_{\text{ph}}$. Both of these discrepancies are attributed to excitons.

21.4 Excitons

When a photon excites an electron into the conduction band, it inevitably leaves behind a hole in the valence band as well. In the context of the single-particle band theory, the energies of these two excitations should simply be added together. However, it is easy to imagine that because the electron and hole are charged particles and because at low temperatures there will not be enough free carriers to screen them, they can lower their energy by approaching one another and binding together. Such bound electron–hole pairs are called *excitons* (Figure 21.7).

The exciton can be considered in two limits. In the first case, the binding is sufficient weak that the hole and electron orbit at a large distance from one another. Then one has a Mott–Wannier exciton. In the second case, the hole and electron are so tightly bound that they sit on the same site. Then one has a Frenkel exciton.

21.4.1 Mott–Wannier Excitons

Excitons are comparatively easy to understand in semiconductors. Because of the large static dielectric constant, the Coulomb interaction between electron and hole is greatly reduced, and their mean separation when bound together is much larger

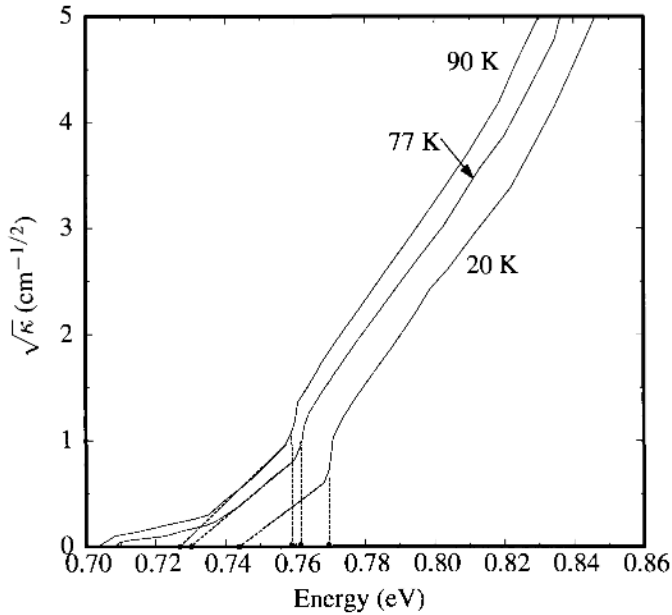


Figure 21.6. Onset of optical absorption in germanium. The square root of the extinction coefficient κ is plotted as a function of incoming photon energy, in the expectation that the experiment will behave according to the prediction of Eq. (21.23). The measurements should look like a straight line, followed by a smooth transition into another straight line of steeper slope. Extrapolation of the first straight line to zero (sloped dotted lines) should give $\mathcal{E}_g - \hbar\omega_{ph}$, while the transition point should give $\mathcal{E}_g + \hbar\omega_{ph}$ (vertical dotted lines). At the very lowest temperature, this prediction is obeyed partly, although there is a cusp at the transition point. At the two higher temperatures, there is also a low-energy tail in the absorption that does not fit the predictions. These discrepancies have been explained quantitatively as consequences of excitons. [Source: Macfarlane et al. (1957), p. 1379.]

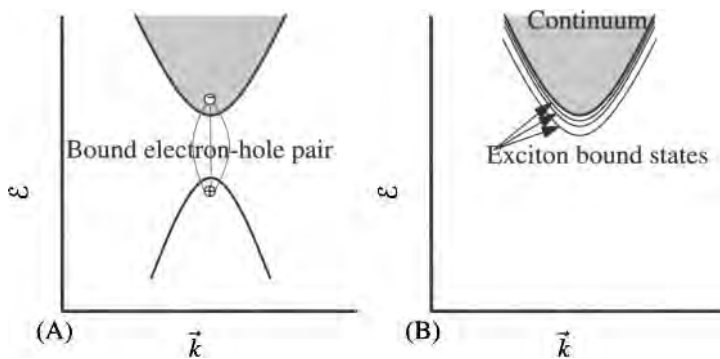


Figure 21.7. Schematic view of energy levels resulting from exciton formation. In (A) an electron and hole bind together. (B) shows the energy levels of the bound pair consisting of an infinite series of discrete levels below the one-electron conduction band that vary with $\vec{k}$ because of their center-of-mass motion, along with a continuum of states above.

than a lattice spacing. In this limit, the exciton is called a *Mott–Wannier exciton*. The same arguments used in Section 18.3.1 to discuss interaction of electrons with weak impurity potentials apply again. Assume that the characteristic distance between electron and hole is much larger than a lattice spacing, so that the wave functions for both electron and hole can be constructed as products of Bloch waves with wave vector $\vec{k}$ near 0. Then the interactions of electron and hole with the periodic ionic potential can be eliminated from the Hamiltonian, provided that their masses are replaced by the appropriate effective masses. That is, the electron and hole obey

$$\left[\frac{-\hbar^2}{2m_n^*} \nabla_{\vec{r}_n}^2 + \frac{-\hbar^2}{2m_p^*} \nabla_{\vec{r}_p}^2 - \frac{e^2}{\epsilon^0 |\vec{r}_n - \vec{r}_p|} - \mathcal{E} \right] \Psi(\vec{r}_n, \vec{r}_p) = 0. \quad \epsilon^0 \text{ is the static dielectric constant.} \quad (21.24)$$

Equation (21.24) is exactly the equation obeyed by the electron and proton in forming the hydrogen atom. It can be separated into two equations, one describing the center-of-mass motion of the exciton and the other describing the binding by defining

$$\vec{R} = \frac{m_n^* \vec{r}_n + m_p^* \vec{r}_p}{m_n^* + m_p^*} \quad (21.25)$$

$$\vec{r} = \vec{r}_n - \vec{r}_p \quad (21.26)$$

to give

$$0 = \left[\frac{-\hbar^2}{2(m_n^* + m_p^*)} \nabla_{\vec{R}}^2 - \mathcal{E}_{\text{cm}} \right] \Psi_{\text{cm}}(\vec{R}) \quad \text{cm = center of mass.} \quad (21.27)$$

$$0 = \left[\frac{-\hbar^2}{2\mu} \nabla_{\vec{r}}^2 - \frac{e^2}{\epsilon^0 r} - \mathcal{E}_b \right] \Psi_b(\vec{r}), \quad \text{b = binding.} \quad (21.28)$$

with the reduced mass μ given by

$$\mu = \frac{m_n^* m_p^*}{m_n^* + m_p^*}. \quad (21.29)$$

There is no need to repeat the solution of the hydrogen atom any further. The bound states of Eq. (21.28) have energies

$$\mathcal{E}_l = -\frac{\mu e^4}{2\hbar^2 \epsilon^0{}^2 l^2} = -\frac{\mu}{m \epsilon^0{}^2 l^2} \cdot 13.6 \text{ eV} \quad l = 1 \dots \infty. \quad (21.30)$$

and are characterized by a radius

$$a_0^* = \frac{\epsilon^0 \hbar^2}{e^2 \mu} = \frac{m \epsilon^0}{\mu} \cdot 0.529 \text{ \AA}. \quad (21.31)$$

Because μ is typically half an electron mass or less, while ϵ for a semiconductor may be around 10, the scale on which the bound state lives is 20 times larger than the usual hydrogen atom; therefore, the picture of the electron and hole sitting in a classical background dielectric is consistent. Binding energies are on the order of a

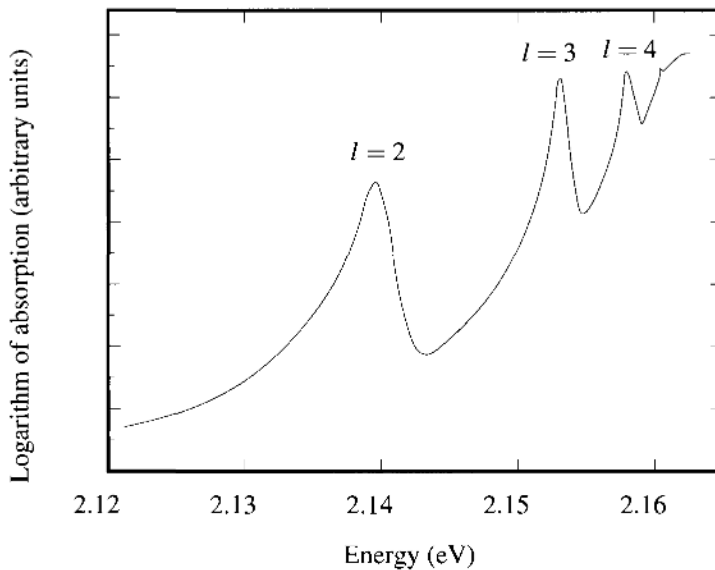


Figure 21.8. Absorption measured in Cu_2O . Cuprous oxide is an indirect gap semiconductor, and absorption begins to increase as a square root, at an energy of 2.045 eV. This square root form is predicted by Elliott (1957) for phonon-assisted transitions to the lowest-lying exciton state; direct transitions are forbidden by symmetry. At about 0.1 eV higher in energy a series of peaks corresponding to electron transitions between the top of the valence band and excited exciton states begins. The first ($l = 2$), second, and third excited states are clearly visible. [Experiments performed at 77 K by Baumeister (1961), p 361.]

hundredth of a Rydberg, or 0.1 eV. The dispersion relations of excitons have been measured with great accuracy, as reported by Ueta et al. (1986).

According to Eq. (21.30), the presence of excitons produces a sequence of discrete energy states sitting near the conduction band edge. Usually these peaks are not resolved individually, but merge into a bump centered around $\hbar\omega = \mathcal{E}_g$. However, at low temperatures in Cu_2O , a sequence of distinct absorption peaks can be measured, as shown in Figure 21.8. In these measurements, an electron makes a transition from the valence band to various excited states of the excitons. In addition, just like the hydrogen atom, Eq. (21.28) must have a continuous collection of unbound solutions, which enhance the absorption for $\hbar\omega > \mathcal{E}_g$. Why are such effects observed in GaAs (Figure 21.5) but not InSb (Figure 21.4)? The static dielectric constant of InSb, recorded in Table 22.2, is 17.88, and the effective mass m^* is $0.01m$, reducing the energy scale of the excitons to $3 \cdot 10^{-4}$ eV and making them invisible even at temperatures as low as 5 K.

21.4.2 Frenkel Excitons

The *Frenkel exciton* consists of a bound state of electrons and holes in insulators, where the spatial extent of the exciton is on the scale of angstroms, and the continuum approximation that served the Mott–Wannier exciton fails. These excitons

have excited states, just like more conventional molecules, and transitions between these excited states allow optical absorption in insulators. In addition, these excitons are mobile and can travel through the lattice as particles. A calculation of their properties, using the Hartree-Fock approximation, appears as Problems 4 and 5.

21.4.3 Electron-Hole Liquid

When a sufficiently large density of excitons is created in a semiconductor, by strong illumination, the excitons can condense into a liquid-like state that manifests itself as an *electron-hole drop*, first predicted by Keldysh and Kozlov (1967) and reviewed by Hensel et al. (1977) and Rice (1977).

21.5 Optoelectronics

Semiconductor electronics can be made to interact with light in numerous different ways, ranging from the conversion of photons into electrical energy to the conversion of electrical energy to laser beams. These various applications are largely implicit in the Kubo formula, Eq. (20.70), but great ingenuity continues to be expended in obtaining efficient performance.

21.5.1 Solar Cells

A solar cell is essentially a diode that has been fabricated so as to absorb as much light as possible in the neighborhood of the junction. Every photon that is absorbed generates an electron-hole pair. The pairs that are generated inside the depletion region are immediately ripped apart by the enormous electric fields and sent to opposite sides. Because the depletion region is very narrow, the formation of pairs in the quasi-neutral regions is more important. Minority carriers formed within a distance L_p or L_n of the depletion zone can diffuse there and are then swept across to the other side of the junction. This picture predicts that shining light of intensity I on a diode will produce a current density

$$j = -I\mathcal{G}(L_p + L_n) \quad \mathcal{G} \text{ is some constant of proportionality taking into account the efficiency of absorption, and making the units come out right.} \quad (21.32)$$

Because the sizes of the quasi-neutral regions are independent of the voltage across the diode, the main effect of light is simply to lower the current-voltage curve of the diode downwards, as shown in Figure 21.9. When the solar cell is under illumination but not connected to a circuit, the separation of electron-hole pairs causes the voltage across it to rise until the current-voltage curve intersects the point where current vanishes, at a voltage on the order of $k_B T/e$, or around 0.025 V. Connected to a load, the voltage drops slightly and current begins to flow, generating power. There is an ideal load at which power generation is maximum, while if the cell is connected across an open circuit the voltage across it drops to zero, and it again generates no power. Solar cells can convert sunlight to power

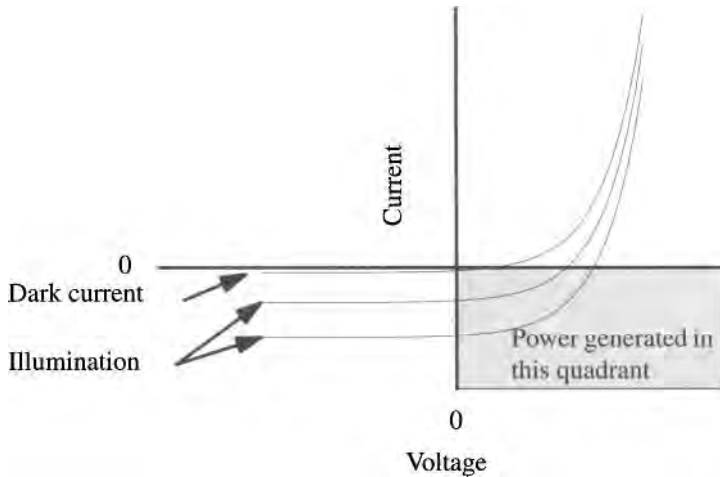


Figure 21.9. The current–voltage characteristic for a solar cell is essentially the ideal diode equation (19.78) shifted vertically downwards. In the lower right quadrant the diode is forward biased, but current flows in the opposite direction, generating power.

with an efficiency of over 25%, but are not yet an economical source of energy under most conditions.

21.5.2 Lasers

The laser is based upon the idea of *stimulated emission*, first conceived by Einstein (1916) and first measured by Ladenburg (1928). Three decades followed the first experimental observations before a coherent beam of radiation was actually produced, and several more decades followed before this “solution in search of a problem” became important commercially. Possible applications of the laser were, however, evident right from the outset.

Hecht (1992) describes the complicated history of the race to complete the laser. G. Gould eventually won extensive patent rights to the laser based upon personal notes written in 1957, but it was an influential paper of Schawlow and Townes (1958) that started the international scramble to build a working device, won by Maiman (1960). The basic idea is fairly simple. If an excited electronic state that should be nearly empty in thermal equilibrium is somehow given a macroscopic occupation of electrons, then as soon as the electrons begin to travel to the ground state the process can proceed explosively. The light emitted by electrons as they jump down the energy levels triggers other electrons to jump down simultaneously, and light emission grows exponentially. To see why, go back to Einstein’s original argument.

Stimulated and Spontaneous Emission. Einstein (1916) deduced stimulated emission from deceptively simple arguments about thermal equilibrium. Consider an isolated cavity filled with material whose index of refraction is $\bar{n}$, and focus upon two electronic states 1 and 2, of energies $\mathcal{E}_1$ and $\mathcal{E}_1 + \mathcal{E}_{12}$. Einstein first ob-

served that all systems prepared in an excited state decay at some rate, and he took the *spontaneous transition rate* R_{sp} from state 2 to 1 to be

$$R_{sp} = A_{21}f_2(1 - f_1) \quad \text{The transition rate is proportional to the probability that state 2 be occupied and that state 1 be empty. } f_2 \text{ and } f_1 \text{ are Fermi factors for the respective states.} \quad (21.33)$$

Next suppose the cavity to be so well isolated that the material inside is in complete equilibrium with a bath of photons at temperature T . Whenever a photon of the right frequency collides with an electron in state 1, it has the chance to be absorbed and bring the electron to state 2. The rate R_{12} at which electrons make transitions from the lower to the upper state is proportional to the probability f_1 that the lower state is occupied, to the probability $(1 - f_2)$ that the upper state is unoccupied, and to the number of photons $N_{\mathcal{E}_{12}}$ of energy $\mathcal{E}_{12}$ in the cavity. As noted in Eq. (20.72), transitions between electronic states are never completely restricted to a single energy, but can be produced by photons in some small range $d\mathcal{E}$. Accordingly, the number of photons producing the transition is proportional to $N_{\mathcal{E}_{12}}D_{ph}(\mathcal{E}_{12})$, where D_{ph} is the energy density of photon states. Because the density of photon states will depend upon the index of refraction of the medium, complicated electron–electron interactions are subtly being included with almost no increase in the complexity of the theory. Thus

$$R_{12} = B_{12}f_1(1 - f_2)N_{\mathcal{E}_{12}}D_{ph}(\mathcal{E}_{12}) \quad B_{12} \text{ is a temperature-independent coefficient to be determined later.} \quad (21.34)$$

The reverse rate, R_{21} , at which an electron jumps down from an excited state, producing photons, must at least be as large as R_{sp} . Einstein guessed that the rate would also rise in proportion to the number of photons in the cavity. Choosing a coefficient B_{21} defined to correspond closely to the coefficient in Eq. (21.34) gives

$$R_{21} = B_{21}f_2(1 - f_1)N_{\mathcal{E}_{12}}D_{ph}(\mathcal{E}_{12}) + A_{21}f_2(1 - f_1). \quad (21.35)$$

The transition rate is the sum of the spontaneous transition rate, independent of the photons, and a term proportional to the photon number.

The contribution proportional to $N_{\mathcal{E}_{12}}$ is the *stimulated emission rate*. Why photons should cause such transitions is not immediately clear, but considering how thermal equilibrium is maintained shows that these transitions must be present.

In thermal equilibrium the population of electrons is stationary, and R_{12} and R_{21} must be equal. Because

$$\frac{f_2(1 - f_1)}{f_1(1 - f_2)} = e^{-\beta\mathcal{E}_{12}}, \quad \text{Just put in Fermi functions to verify. Equivalent to the statement that the probability of occupying any quantum state in equilibrium is proportional to } e^{-\beta\mathcal{E}}. \quad (21.36)$$

one has

$$R_{12} = R_{21} \quad (21.37)$$

$$\Rightarrow B_{12}N_{\mathcal{E}_{12}}D_{ph}(\mathcal{E}_{12}) = e^{-\beta\mathcal{E}_{12}} [B_{21}N_{\mathcal{E}_{12}}D_{ph}(\mathcal{E}_{12}) + A_{21}] \quad (21.38)$$

$$\Rightarrow D_{\text{ph}}(\mathcal{E}_{12})B_{12} - A_{21} = e^{-\beta\mathcal{E}_{12}} [D_{\text{ph}}(\mathcal{E}_{12})B_{21} - A_{21}] \frac{N_{\mathcal{E}_{12}}}{1/(\exp[\beta\mathcal{E}_{12}] - 1)} \quad (21.39)$$

$$\Rightarrow A_{21} = D_{\text{ph}}(\mathcal{E}_{12})B_{21} \text{ and } B_{12} = \quad (21.40)$$

If Eq. (21.39) is to hold independent of temperature.

Equation (21.40) can also be written as

$$R_{21} = B_{21}f_2(1 - f_1)(N_{\mathcal{E}_{12}} + 1)D_{\text{ph}}(\mathcal{E}_{12}). \quad (21.41)$$

In this form, change in electron energies by emission and absorption of photons reveals a perfect parallel with the change in neutron energies by emission and absorption of phonons described in Eq. (13.134). The probability of transitions where electron energy increases is proportional to $N_{\mathcal{E}_{12}}$, while the probability of transitions where it decreases is proportional to $N_{\mathcal{E}_{12}} + 1$, with the trailing factor of 1 responsible for spontaneous emission.

Einstein's argument is completed by asserting that in nonequilibrium situations, as when a stream of light energy hits electrons and is ultimately dispersed as heat, the occupation probabilities f_1 and f_2 need not have the values used to obtain Eq. (21.36), but the transition rates R_{12} and R_{21} will continue to depend upon f_1 and f_2 in the ways indicated by Eqs. (21.34) and (21.41). Thus, away from equilibrium, the absorption of energy by electrons from photons will be proportional to

$$R_{12} - R_{21} = B_{21}[(f_1 - f_2)N_{\mathcal{E}_{12}} - f_2(1 - f_1)]D_{\text{ph}}(\mathcal{E}_{12}). \quad (21.42)$$

The absorption of energy is given by the difference between absorption and emission, as given by Eqs. (21.34) and (21.41).

The coefficient B_{12} is the only remaining unknown of the theory, and it can be determined by comparison with the Kubo–Greenwood formula. When the photon density $N_{\mathcal{E}_{12}}$ is very large, spontaneous emission becomes negligible, and Eq. (21.42) must coincide with the results of the semiclassical theory of absorption. Therefore, turn to Eq. (20.71), which predicts that if one is considering only the the two quantum states 1 and 2, then the conductivity is

$$\text{Re}[\sigma_{\alpha\beta}(\omega)] = \frac{e^2\pi}{\hbar\omega m^2\mathcal{V}}(f_1 - f_2)F_{12}(\omega)\langle 1|\hat{P}_\alpha|2\rangle\langle 2|\hat{P}_\beta|1\rangle. \quad (21.43)$$

$F_{12}(\omega)$ is the lineshape, a function with unit area, and peaked at $\omega_2 - \omega_1$.

In practical situations, there will not simply be two levels 1 and 2, but instead a large number N of atoms or impurities that can be excited from one level to another. Summing over these many atoms will have the effect of averaging over all possible spatial orientations of the excited state $|2\rangle$ and replacing the matrix elements by their spatial average $\sum_\beta |\langle 1|\hat{P}_\beta|2\rangle|^2/3$. Using Eqs. (20.14) and (20.21) to relate conductivity to the attenuation constant $\alpha = -g$ shows that radiation intensity I grows as $\exp[gx]$ where the *gain* is

$$g(\omega) = \frac{N}{\mathcal{V}} \frac{4\pi^2 c^2}{\omega \bar{n}} \left(\frac{e^2}{\hbar c} \right) F_{12}(\omega)(f_2 - f_1) \frac{\sum_\beta |\langle 1|\hat{P}_\beta|2\rangle|^2}{3m^2 c^2}. \quad (21.44)$$

The net rate $R_{21} - R_{12}$ at which electrons travel between states 2 and 1 equals the rate at which photons are produced per time. To find the total number of photons, multiply g in Eq. (21.44) by $c/\bar{n}$ to find the time rate of change of light intensity I . Next recall that light intensity is proportional to photon number through $I = N_{\mathcal{E}_{12}} \hbar \omega$, and finally that in a medium with macroscopic index of refraction $\bar{n}$, the density of photon states per volume is

$$D_{\text{ph}}(\omega) = \frac{\bar{n}^3 \omega^2}{\pi^2 c^3}. \quad (21.45)$$

This is density of states per volume; a factor of $\mathcal{V}$ is needed to find total numbers of photons.

Multiplying Eq. (21.44) by Eq. (21.45), multiplying by $\mathcal{V}c/\bar{n}$, and integrating over ω gives

$$R_{21} - R_{12} = \frac{\partial}{\partial t} N_{\mathcal{E}_{12}} = -N \frac{\mathcal{E}_{12}}{\hbar} \bar{n} \left(\frac{e^2}{\hbar c} \right) 4(f_1 - f_2) \sum_{\beta} \frac{|\langle 1 | \hat{P}_{\beta} | 2 \rangle|^2}{3m^2 c^2} N_{\mathcal{E}_{12}} \quad (21.46)$$

N is the number of host atoms or impurity sites, while $N_{\mathcal{E}_{12}}$ is the number of photons. B_{12} can be determined from Eqs. (21.42) and (21.45) if desired, but this rate equation for photons is more useful than B_{12} . The lineshape $F_{12}(\omega)$ integrates to 1.

Equation (21.44) contains a theoretical motivation for building a laser. If it is possible to construct an initial state in which $f_2 > f_1$, then the attenuation will be negative, and light will grow exponentially as it passes through the medium until state 2 is depleted. Under these circumstances, $-\alpha = g$ as computed in Eq. (21.44) is renamed the gain, and the symbol α is reserved for sources of attenuation and loss that compete with the growth from population inversion.

In order to obtain a laser, it is not enough for the intensity of a beam of light passed into the system to grow. It must be possible for external light sources to be turned off, while light emission from the system continues to grow spontaneously. This condition can be achieved by placing material in a cavity with reflecting ends, as shown in Figure 21.10. If the reflection coefficient of the two ends is $\mathcal{R}$, then the condition for amplification in the laser cavity is

$$\mathcal{R} \exp[x(g - \alpha)] > 1. \quad (21.47)$$

Light has to be able to make a round trip in the cavity and end up stronger than at the beginning. The gain g due to stimulated emission competes with loss processes summarized in the coefficient α .

Pumping. The discussion has hinged upon the assumption of *population inversion*: that an excited state has been populated so that $f_2 > f_1$. Occupying an excited state preferentially at the expense of one lying lower in energy is easy to describe, but violates conditions of thermal equilibrium, so it is not immediately obvious how to achieve it in practice.

Lasers can be created in numerous different ways and in many different types of materials, including liquids and gases. The following discussion will focus upon *solid-state lasers*, which does not refer exclusively to lasers formed of semiconductors, but includes any case in which the laser material is solid. The most important ideas are *metastable states* and *pumping*. For the purposes of laser physics,

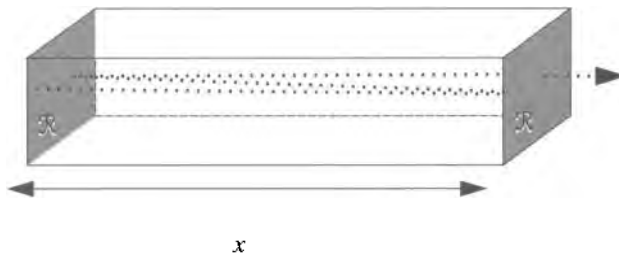


Figure 21.10. Light in a laser cavity reflects several times back and forth from mirrored ends of reflection coefficient $\mathcal{R}$ so as to stimulate more light emission before exiting.

a metastable state is an excited energy level with a lifetime on the order of 10^{-5} to 10^{-3} s. Characteristic *non-radiative decay* times of excited states in solids are usually on the order 10^{-8} to 10^{-11} s, because this is the time that phonons typically require in order to transfer energy from an excited level. A first requirement of a solid-state laser is therefore to find an excited state that cannot be depleted by single phonons, either because available transitions exceed the maximum phonon frequency or because symmetry causes matrix elements [analogous to the ones in Eqs. (13.99) or (20.70)] between initial and final states to vanish. Transitions driven by multiple phonons are still possible, but sufficiently unlikely that emission of photons has a chance to win.

A metastable state has a long lifetime, and by the energy uncertainty principle it can only be excited from the ground state by energy lying in a very narrow range. Because only a laser can produce light with such a narrow linewidth, it would seem that a laser is required to populate the levels upon which laser action is to rely. This difficulty is overcome by pumping, illustrated in Figure 21.11. Rather than occupying the metastable state directly, one pushes electrons into one or more states with short lifetimes that lie above the desired metastable state, and that rapidly decay to it in non-radiative fashion.

Solid-State Ionic Lasers. One way to make a laser is by employing the discrete energy levels provided by color centers in insulating crystals (Section 22.4). The first laser, built from a crystal of ruby by Maiman (1960), was a *three-state* laser involving pump states and a metastable state decaying to the ground state, as shown in Figure 21.12(A). In order to occupy the 2E state upon which the laser relied, the crystal was illuminated with a powerful flashlamp that succeeded in placing more electrons into 2E than were left in the ground state. Numerous researchers realized that a *four-state* laser would have considerable advantages because if the state at the bottom of the laser transition was normally unoccupied, constraints upon the occupation of the upper level become less severe. Three-state lasers typically operate in pulsed fashion, with bursts of laser radiation following flashes of pumping light, while four-state lasers can operate continuously. The energy levels of Nd in yttrium aluminum garnet (YAG), the impurity in one of the most important solid-state lasers appear in Figure 21.12(B).

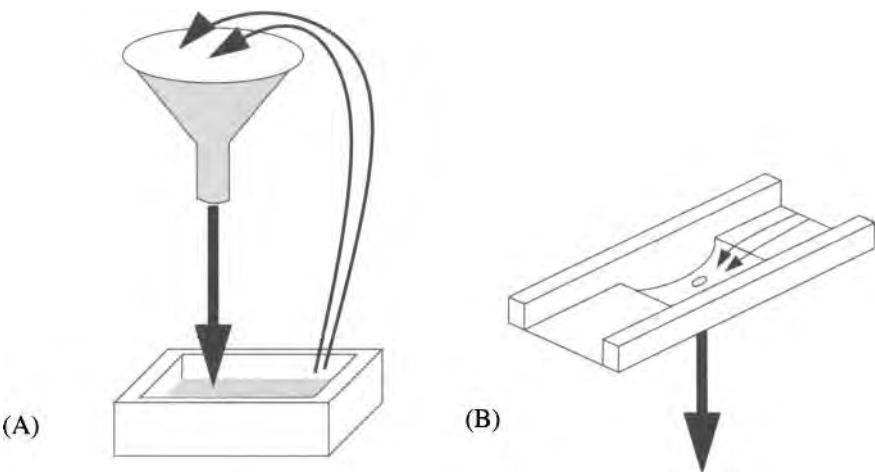


Figure 21.11. (A) The problem of occupying a metastable atomic state is a bit like pumping water from a trough into a funnel. The pumping action can be sloppy and erratic, but the water nevertheless exits the funnel in a narrow steady beam. (B) For a semiconductor laser, the pumping scheme is a bit different. As electrons travel along, they pass into a region where the doping has changed, and the Fermi level drops precipitously from conduction to valence band, leading to a laser transition somewhat like water falling through a drain. The double heterostructure laser ensures that the optical transitions take place in a confined spatial region by trapping the electrons with an additional slight drop in Fermi level of the conduction band that resembles a basin.

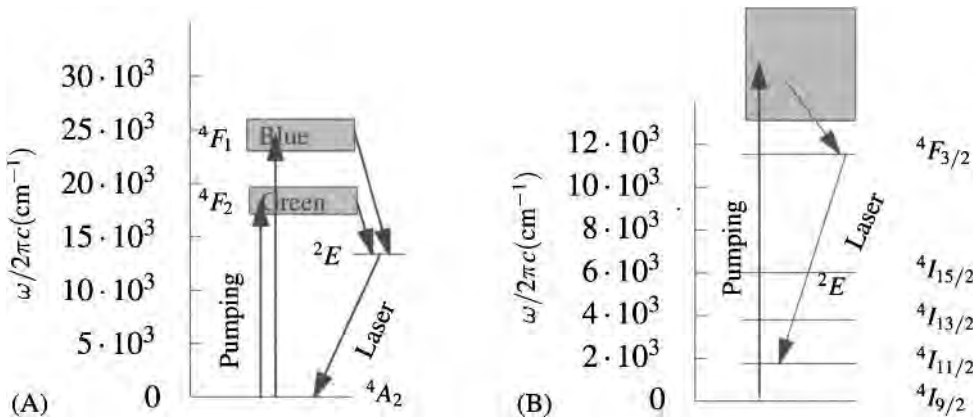


Figure 21.12. (A) Energy levels of Cr³⁺ in Al₂O₃ (ruby). Electrons are pumped to high levels, and they quickly decay to metastable state 2E . The laser transition then occurs to 4A_2 . (B) Energy levels of Nd in Y₃Al₅O₁₂ (Nd:YAG). The metastable state is $^4F_{3/2}$, and the laser transition occurs to state $^4I_{9/2}$. Transitions to other states also occur, but the transition to $^4I_{11/2}$ is the most likely, at 60% probability. Energy levels are being described in LS coupling, as described by Schiff (1968), p. 435, or Landau and Lifshitz (1977), p. 250, where the superscript gives $2S + 1$, the subscript gives J , and the capital letter indicates L .

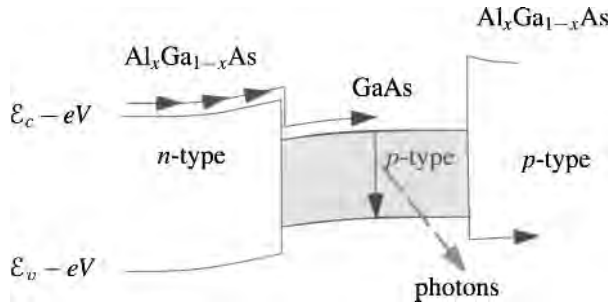


Figure 21.13. A double heterojunction structure makes use of two junctions to trap electrons and force them into radiative transitions in a small region. Electrons arriving from the n -doped region on the left find themselves trapped in a p -type region and hemmed in on both sides. The central layer also has a higher index of refraction than the surrounding regions, confining light in a cavity.

Light-Emitting Diodes and Semiconductor Lasers. The light-emitting diode and semiconductor laser exploit the fact that electrons in n -type material traveling in the conduction band must rapidly fall to the valence band after entering p -type material. In the process, they emit photons whose energy equals the band gap. The light-emitting diode simply extracts this radiation. To obtain a laser, it is necessary to obtain a high spatial density of recombining electrons and holes, and to trap the light they emit in a cavity. Practical devices are based upon the double heterojunction structure shown in Figure 21.13. The central region acts like a trap where electrons collect in large numbers and recombine with high efficiency.

If an indirect gap semiconductor such as silicon were used in this application, electrons would collect at the lowest point in the conduction band but then find themselves unable to make the transition to the valence band without assistance from phonons. The efficiency of the device would be unacceptable. That is why direct gap semiconductors such as GaAs are needed for optical applications.

Problems

1. Optical measurements:

- Use the locations of the peaks for light and heavy holes to estimate the frequency ω at which the experiment in Figure 21.3 was conducted.
- The data in Figure 21.6 allow one to estimate $\omega_{\text{ph}}(\vec{\delta}k)$. Using the data at 20 K, what are ω_{ph} and $\vec{\delta}k$?

- Lifetime of hydrogen excited state:** Many of the quantities appearing in the theory of lasers can be evaluated explicitly for hydrogen. Consider transitions between the 2P and 1S states, using the wave functions

$$\psi_1 = \frac{\exp[-r/a_0]}{\sqrt{\pi a_0^3}} \quad \text{and} \quad \psi_2 = \frac{x \exp[-r/2a_0]}{\sqrt{2^5 \pi a_0^5}}. \quad (21.48)$$

- (a) Evaluate the integral $P_{12} \equiv \langle 1 | \hat{p}_x | 2 \rangle$.
- (b) Evaluate ω for the $2P \rightarrow 1S$ transition.
- (c) How must Eq. (21.44) be generalized to account for the sixfold degeneracy of the $2P$ state?
- (d) Show that in a box with N hydrogen atoms, the transition rate A_{21} from one of the six $2P$ states to one of the two $1S$ states is given by

$$A_{21}(\omega)(f_2 - f_1) = \frac{1}{6N} g(\omega) \frac{c}{\bar{n}}. \quad (21.49)$$

- (e) Set $\bar{n} = 1$, and find A_{21} by integrating $A_{21}(\omega) D_{\text{ph}}(\omega)$ over ω .
- (f) Put A_{21} into a final form involving e , m , $\hbar$, and c . The numerical answer is $6.268 \cdot 10^8 \text{ s}^{-1}$.

3. **Gunn effect:** The conduction band of gallium arsenide has the form shown in Figure 19.8. In the central minimum, electrons have a small effective mass ($m^* = 0.07m$), leading to a mobility on the order of $7500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. However, once the electron acquires a $\vec{k}$ in the minimum on the right, its effective mass increases to around $m^* = 0.4m$, and its mobility decreases to $200 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. The result is that electron velocity as a function of electric field E looks as shown in Figure 21.14.

When a sufficiently large electrical field ($\sim 3000 \text{ V/cm}$) is applied to GaAs, the crystal begins to emit pulses, typically at frequencies in the 50 GHz range. The physical origin of the pulses lies in the fact that as electrons become more energetic, it is favorable for them to jump from the center valley at Γ to the left valley at L in Figure 19.8. However, when they do so, their mobility decreases, and they slow down. This leads to a traffic jam of charge, which travels through the sample as a localized pulse. The goal of this problem is to calculate properties of these moving pulses.

Model electron velocity versus field by

$$v(E) = \begin{cases} \mu_1 E & \text{for } E < E_c \\ \mu_2 E & \text{for } E > E_c, \end{cases} \quad (21.50)$$

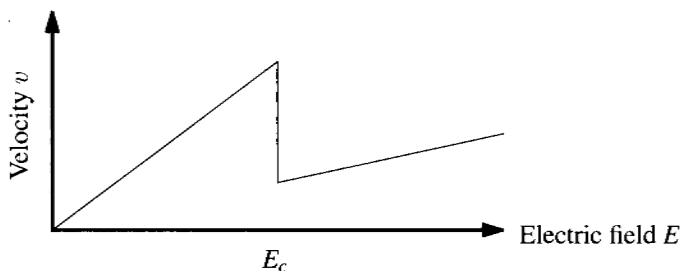


Figure 21.14. Schematic view of velocity as a function of field for electrons in gallium arsenide. Electron mobility decreases for fields greater than the critical E_c .

with $\mu_1 > \mu_2$, as shown in Figure 21.14. Denote net charge density as a function of length x along the sample by $n(x)$; $n(x)$ vanishes far from the pulse. Denote current by j , and take it to be of the form

$$j = nev(E) - eD \frac{\partial n}{\partial x}, \quad (21.51)$$

where e is the electron charge and D is a diffusion constant.

- (a) Write down Poisson's equation for the electric field, and an equation relating the current j and the time rate of change of the charge density n . In order to describe a pulse moving at velocity v^* , assume that all spatially varying fields are moving in steady state at velocity v^* , and use this assumption to simplify the relation between current and charge density.
- (b) Look for a solution such that charge density vanishes for large negative x and large positive x . Assume that for large negative x the electric field approaches a low value $E_1 < E_c$, and for large positive x it approaches a larger value $E_2 > E_c$. Find an expression for $\partial n / \partial E$, and integrate it to find charge as a function of electric field.
- (c) Assume that

$$\mu_1 E_1 = \mu_2 E_2 = v^*. \quad (21.52)$$

Adding to these equations the condition that charge density vanish when the field reaches E_2 , find v^* in terms of the mobilities μ_1, μ_2 , and E_c .

- (d) Find the electric field $E(x)$ for $E < E_c$.
- (e) As a pulse of this type moves across a sample, what happens to the potential drop from one end of the sample to the other? Suppose that one demands that the potential drop across the sample remain constant as the pulse moves in its interior. Draw a sketch indicating how one could construct solutions that would make it possible to satisfy this condition.

4. **Frenkel exciton I:** To model the Frenkel exciton, consider a variant of the tight-binding model. At each site of a lattice there is an electron, which can occupy one of two states, an atomic ground state and an atomic excited state. Denote the Wannier states by $w_g(\vec{r} - \vec{R})$ and $w_e(\vec{r} - \vec{R})$; suppose they are real functions and are localized near the atomic sites.

When the atoms are assembled into a crystal, the ground state is given to good approximation by a Hartree–Fock state, consisting of an antisymmetrized product for N electrons of $w_g(\vec{r}_i - \vec{R})$, the product being taken over all N sites in the lattice. In second-quantized notation, the ground state of the system is

$$|\Phi_0\rangle = \prod_{\vec{R}} \hat{c}_{\vec{R},g}^\dagger |0\rangle. \quad \hat{c}_{\vec{R},g}^\dagger \text{ creates an electron in the atomic ground state at site } \vec{R}. \quad (21.53)$$

Suppose now that the only quantum states of importance are the ground state and the excited states where one electron at one site moves from the ground to the excited state; such a state is

$$|\vec{R}\rangle \equiv \hat{c}_{\vec{R}e}^\dagger \hat{c}_{\vec{R}g} |\Phi_0\rangle. \quad (21.54)$$

- (a) Choose the zero of energy so that $\langle \vec{R} | \hat{\mathcal{H}} | \Phi_0 \rangle = 0$. If the states in Eqs. (21.53) and (21.54) are all that matter, then the Hamiltonian is given by

$$\hat{\mathcal{H}} = |\Phi_0\rangle \langle \Phi_0 | \hat{\mathcal{H}} | \Phi_0 \rangle \langle \Phi_0 | + \sum_{\vec{R}\vec{R}'} |\vec{R}\rangle \langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle \langle \vec{R}' |. \quad (21.55)$$

Show that

$$|\Phi_k\rangle = \frac{1}{\sqrt{N}} \sum_{\vec{R}} e^{i\vec{k}\cdot\vec{R}} |\vec{R}\rangle \quad (21.56)$$

is an eigenstate, and find the eigenvalue. Use the fact that $\langle \vec{R} + \vec{R}' | \hat{\mathcal{H}} | \vec{R} \rangle$ depends only upon $\vec{R}'$.

- (b) Now proceed to evaluate the matrix elements $\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle$ that determine this eigenvalue. The Hamiltonian $\hat{\mathcal{H}}$ contains kinetic energy and Coulomb interaction terms. There are many contributions to $\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle$; however, this matrix element causes something proportional to the unit operator to be added to the Hamiltonian, is equivalent to a constant shift in energy, and can be neglected. Therefore, focus on evaluating $\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle$ for $\vec{R} \neq \vec{R}'$. Assuming that Coulomb interaction leads to the largest interaction between sites, take

$$\hat{\mathcal{H}} = \frac{e^2}{2} \sum_{l \neq l'} \frac{1}{|\vec{r}_l - \vec{r}_{l'}|}. \quad \text{This is the Schrödinger representation; do not} \quad (21.57)$$

confuse the independent variable $\vec{r}_l$ with the
wave function centered at $\vec{R}$, which is denoted
by $|\vec{R}\rangle$.

Write the Hamiltonian (21.57) in second quantized notation, using Eq. (C.10).

- (c) Show that

$$\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle = e^2 \int \frac{d\vec{r}_1 d\vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} \left[w_g(\vec{r}_1 - \vec{R}') w_e(\vec{r}_2 - \vec{R}) w_e(\vec{r}_1 - \vec{R}') w_g(\vec{r}_2 - \vec{R}) \right. \\ \left. - w_e(\vec{r}_1 - \vec{R}) w_g(\vec{r}_2 - \vec{R}') w_e(\vec{r}_1 - \vec{R}') w_g(\vec{r}_2 - \vec{R}) \right]. \quad (21.58)$$

- (d) Argue from localization of the wave functions that the second term in Eq. (21.58) can be neglected, and

$$\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle \approx e^2 \int \frac{d\vec{r}_1 d\vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} w_g(\vec{r}_1 - \vec{R}') w_e(\vec{r}_2 - \vec{R}) w_e(\vec{r}_1 - \vec{R}') w_g(\vec{r}_2 - \vec{R}). \quad (21.59)$$

5. Frenkel exciton II:

- (a) Continuing the study of the Frenkel exciton, let $\vec{\delta R} \equiv \vec{R} - \vec{R}'$, $\delta \vec{r}_1 = \vec{r}_1 - \vec{R}'$ and $\delta \vec{r}_2 = \vec{r}_2 - \vec{R}$. Show that

$$\frac{\delta R}{|\vec{r}_1 - \vec{r}_2|} \approx 1 + \frac{1}{2} \left[\frac{2(\delta \vec{r}_1 - \delta \vec{r}_2) \cdot \vec{\delta R}}{\delta R^2} - \frac{(\delta \vec{r}_1 - \delta \vec{r}_2)^2}{\delta R^2} \right] + \frac{3}{8} \left[\frac{2(\delta \vec{r}_1 - \delta \vec{r}_2) \cdot \vec{\delta R}}{\delta R^2} \right]^2 \dots \quad (21.60)$$

- (b) Define the dipole moment

$$\vec{p} = e \int d\vec{r} w_g(\vec{r}) w_e(\vec{r}) \vec{r}. \quad (21.61)$$

Show that

$$\langle \vec{R} | \hat{\mathcal{H}} | \vec{R}' \rangle \approx \frac{p^2}{\delta R^3} - 3 \frac{(\vec{p} \cdot \vec{\delta R})^2}{\delta R^5} \quad (21.62)$$

$$= -\vec{p} \cdot \left[\vec{\nabla} \left(\vec{p} \cdot \vec{\nabla} \frac{1}{\delta R} \right) \right]. \quad (21.63)$$

- (c) Using Eq. (21.63), find the energy of state $\Phi_{\vec{k}}$. Convert sums over $\delta \vec{R}$ to integrals to find the final expression.

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